

listed programming languages that showed a positive growth in all the four ranking schemes. There were just two — Python and JavaScript. This time the number has gone down further. The only programming language with a positive growth rate is Python. In fact, Python is one of the top three programming languages in all the four rankings. So, if you are learning programming in 2021 and want to make a career out of it, my suggestion is to learn Python. I don't want to contradict myself but there could be a downside to learning Python too, as a large number of people will be learning this language. So we need to explore other options also. Though they do not have the support of a very large programming community like Python or JavaScript, I would like to mention two other programming languages from our top 10 list as winners — R and TypeScript. The popularity of these two programming languages has been steadily increasing over the years and they might become a permanent fixture in the top 10 list in all the programming language rankings in the near future itself. Figure 1 shows the logo of R and Figure 2 shows the logo of TypeScript. Unlike Python, which has a universal appeal, these two languages are restricted in their application domains. But I still believe you could make a decent career out of learning any one of these languages. Please don't assume that I am in favour of learning a scripting language like TypeScript, or a language like R which is mostly used for statistical computing as a person's first programming language. No! I am not. Python or C could be a suitable first programming language. But, at a later point, you could surely give R or TypeScript a try.



Figure 1: The logo of R



Figure 2: The logo of TypeScript

Now let us see which programming languages are steadily losing their ground. In 2019, after consulting all the four rankings, I opined that Ruby and Perl are going down in popularity. But I am not so pessimistic about Ruby now. As it is still ranked 7th, 11th, 15th, and 15th in the four rankings, I don't have the courage to say any more that Ruby may not have a bright future. But I still am pessimistic about Perl. It is ranked 17th, 18th, 23rd, and 30th in the four rankings. Moreover, in October 2019, Perl 6 was officially renamed as Raku and is now considered a programming language that is independent of Perl (Perl 5). So, I believe that if anybody wants to learn Perl newly in 2021, they should consult an expert and make sure that they have made the right choice. Though ranked 8th, 11th, 18th, and 28th in the four rankings, I have doubts about the future of Objective-C also. The rising popularity of Swift, which is similar in spirit to Objective-C and targets the same audience — mostly Apple iOS and macOS developers — is the reason for my scepticism about the future of Objective-C. So, I believe, if somebody is going to learn this language afresh in 2021, they should consult an expert and check if learning Swift is more beneficial in the long run. Of course, I could be entirely wrong about Perl and Objective-C and might be eating my words, again, the next time I write about the popularity of programming languages.

The young contenders in the programming world revisited

In 2019, we had made a list of seven promising programming languages that were first released in the past decade and were listed in the top 50 in at least one of the four rankings. Two years have passed, and let us revisit these programming languages and see if they are as promising now as they were in 2019. The programming languages in the list then were Swift, Go, Rust, Kotlin, Julia, TypeScript, and Dart.

With the initial version released in June 2014 and being a permanent fixture in our top 10 list of popular programming languages for the years 2017, 2019, and 2021, there is no further need to justify the inclusion of Swift in this list. TypeScript, whose initial version was released in October 2012, has entered our top 10 list for the first time in 2021 and hence the inclusion in this list too is justified. With one top 10 and three other top 20 finishes in the four rankings, and an overall increase in the ratings in all the four rankings, Go (first released in November 2009) also remains a very promising programming language. Rust has three top 20 finishes and still looks promising. Kotlin also has three top 20 finishes. Though Dart has just one top 20 finish, it has two other top 25 finishes and the ratings are increasing, though marginally. With the support of Google, Dart might also become more relevant in the future. Julia also has one top 20 finish and two other top 25 finishes. But being mainly used for scientific computing and numerical analysis, Julia stands apart from the other promising languages and hence is still very significant.

Thus, all the languages in our 2019 list are still significant and popular. But Swift, TypeScript, and Go have clearly moved ahead of the pack. Now comes the important question: Are there any new programming languages that must be included in this list? Yes! There is one. Solidity, a blockchain programming language first proposed in August 2014, is ranked 48th in the TIOBE Index. With blockchain technologies like smart contracts becoming more popular, I believe Solidity will steadily improve its ranking in the coming years. Hence, the eight programming languages in our list of promising programming languages for the year 2021 are: Swift, TypeScript, Go, Rust, Kotlin, Dart, Julia and Solidity.